

Assignment # 3 – Solution

1. (a) (i) Solution:

Suppose $a|b$ and $a|c$. Then there are integers m and n such that $b = am$ and $c = an$.
Then $b + c = am + an = a(m+n)$, so $a|(b + c)$.

(ii) Solution:

Suppose $a|b$. Then there is an integer n such that $b = an$.
Then $bc = (an)c = a(nc)$, so $a|(bc)$.

(iii) Solution:

If $a|b$ and $b|c$. Then there are integers m and n such that $b = am$ and $c = bn$.
Then $c = bn = (am)n = a(mn)$, so $a|c$.

(b) Solution:

In any two consecutive integers.

one of them will be even ($2m$) and the other will be odd ($2m + 1$)
 $= 2m \times (2m + 1) = 2(2m^2 + m)$. Hence the product is even.

(c) Solution:

Given two odd integers $2m + 1$ and $2n + 1$ and an even integer is $2k$.
 $= (2k)(2m + 1)(2n + 1) = 2(k(2m + 1)(2n + 1))$. Hence the product is an even integer.

2. (a) Solution:

Let $\sqrt{a + b} = \sqrt{a} + \sqrt{b}$

Squaring, we get $a + b = a + b + 2\sqrt{a}\sqrt{b}$

$$\Rightarrow 0 = 2\sqrt{a}\sqrt{b} \quad \text{cancelling } a + b$$

$$\Rightarrow 0 = 2\sqrt{ab}$$

$$\Rightarrow 0 = ab \quad \text{squaring}$$

$$\Rightarrow \text{either } a = 0 \text{ or } b = 0$$

It means that if we want to find out the integers which satisfy the given condition then one of them must be zero.

Hence if we let $a = 0$ and $b = 3$ then

$$\text{R.H.S} = \sqrt{a + b} = \sqrt{0 + 3} = \sqrt{3}$$

Now,

$$\text{L.H.S} = \sqrt{0} + \sqrt{3} = \sqrt{3}$$

From above it is quite clear that the given condition is satisfied if we take $a=0$ and $b=3$.

(b) Solution:

1. Assume r is a rational number. By the definition of rational numbers, r can be written as $\frac{a}{b}$, where a and b are integers and $b \neq 0$.

$$r = \frac{a}{b}$$

2. Now, consider the square of r :

$$r^2 = \left(\frac{a}{b}\right)^2 = \frac{a^2}{b^2}$$

3. Since the square of an integer is still an integer, a^2 and b^2 are integers. Additionally, $b^2 \neq 0$ because $b \neq 0$.

4. Therefore, $r^2 = \frac{a^2}{b^2}$ is a ratio of two integers with a non-zero denominator, which means r^2 is a rational number.

(c) Solution:

Here we are asked to show a single integer for which $2^n - 1$ is prime. First of all, we will check the integers from 1 and check whether the answer is prime or not by putting these values in $2^n - 1$. When we got the answer is prime then we will stop our process of checking the integers and we note that,

Let $n = 7$, then

$$2^n - 1 = 2^7 - 1 = 128 - 1 = 127 \text{ and we know that } 127 \text{ is prime.}$$

3. (a) Solution:

Suppose n and m are odd and $n + m$ is not even (odd i.e., by taking contradiction).

Now $n = 2p + 1$ for some integer p and $m = 2q + 1$ for some integer q

$$\text{Hence } n + m = (2p + 1) + (2q + 1) = 2p + 2q + 2 = 2(p + q + 1)$$

which is even, contradicting the assumption that $n + m$ is odd.

(b) Solution:

Suppose there exists an integer a and a prime number p such that $p|a$ and $p|(a+1)$.

Then by definition of divisibility there exist integer r and s so that

$$a = p \cdot r \text{ and } a + 1 = p \cdot s$$

It follows that

$$1 = (a + 1) - a$$

$$= p \cdot s - p \cdot r$$

$$= p \cdot (s - r) \quad \text{where } s - r \in \mathbb{Z}$$

This implies $p | 1$.

But the only integer divisors of 1 are 1 and -1 and since p is prime $p > 1$. This is a contradiction. Hence the supposition is false, and the given statement is true.

(c) Solution:

Suppose the set of prime numbers is finite.

Then, all the prime numbers can be listed, say, in ascending order:

$$p_1 = 2, p_2 = 3, p_3 = 5, p_4 = 7, \dots, p_n$$

Consider the integer

$$N = p_1 \cdot p_2 \cdot p_3 \cdot \dots \cdot p_n + 1$$

Then $N > 1$. Since any integer greater than 1 is divisible by some prime number p , therefore $p | N$.

Also since p is prime, p must equal one of the prime numbers

$$p_1, p_2, p_3, \dots, p_n$$

Thus

$$p | (p_1 \cdot p_2 \cdot p_3 \cdot \dots \cdot p_n)$$

But then

$$p \nmid (p_1 \cdot p_2 \cdot p_3 \cdot \dots \cdot p_n + 1)$$

$$\text{So } p \nmid N$$

Thus $p | N$ and $p \nmid N$, which is a contradiction.

Hence the supposition is false and the theorem is true.

4. (a) Solution:

The contrapositive statement is:

if $x \leq 1$ and $x \geq -1$ then $|x| \leq 1$ for $x \in \mathbb{R}$.

Suppose that $x \leq 1$ and $x \geq -1$

$$\Rightarrow x \leq 1 \text{ and } x \geq -1 \quad \Rightarrow -1 \leq x \leq 1$$

and so $|x| \leq 1$ Equivalently $|x| > 1$.

(b) Solution:

"For all integers m and n , if m and n are not both even and m and n are not both odd, then $m + n$ is not even."

Or more simply, "For all integers m and n , if one of m and n is even and the other is odd, then $m + n$ is odd"

Suppose m is even and n is odd. Then

$$m = 2p \text{ for some integer } p \quad \text{and} \quad n = 2q + 1 \quad \text{for some integer } q$$

$$\text{Now } m + n = (2p) + (2q + 1) = 2 \cdot (p + q) + 1 = 2 \cdot r + 1 \quad \text{where } r = p + q \text{ is an integer. Hence } m + n \text{ is odd.}$$

Similarly, taking m as odd and n even, we again arrive at the result that $m + n$ is odd. Thus, the contrapositive statement is true. Since an implication is logically equivalent to its contrapositive so the given implication is true.

(c) Solution:

Contraposition: if a and b are positive, even integers, then ab is divisible by 4.

Suppose that $a = 2k$ and $b = 2k$. Then, we have

$$ab = (2k)(2k) = 4k^2$$

Hence, ab is divisible by 4.

5. (a) (i) Solution: Let the prime number n be 7, then $n + 2 = 7 + 2 = 9$ which is not prime.

(ii) Solution: Consider the even number 6.

$$6 \div 4 = 1.5$$

Since 6 is not divisible by 4, this disproves the statement.

(b) (i) Solution:

Consider the positive integer $x = 4$. 4 is not a prime number since it has divisors other than 1 and itself (i.e., 2).

(ii) Solution: For all real numbers x , $x^2 > x$.

Counterexample: Consider the real number $x = 0$.

$x^2 = 0^2 = 0$ and $x = 0$. Here, $x^2 = 0$ is not greater than $x = 0$. This disproves the statement.

Similarly, consider $x = 1$:

$x^2 = 1^2 = 1$ and $x = 1$. Again, $x^2 = 1$ is not greater than $x = 1$.

(c) Solution:

We know that $\sqrt{2}$ is an irrational number. Now $(\sqrt{2})(\sqrt{2}) = \frac{2}{1}$ which is a rational number. Hence the statement is disproved.

6. (a) Solution:

Suppose $6 - 7\sqrt{2}$ is rational.

Then by definition of rational,

$$6 - 7\sqrt{2} = \frac{a}{b}$$

for some integers a and b with $b \neq 0$.

Now consider,

$$7\sqrt{2} = 6 - \frac{a}{b}$$

$$\Rightarrow 7\sqrt{2} = \frac{6b - a}{b}$$

$$\Rightarrow \sqrt{2} = \frac{6b - a}{7b}$$

Since a and b are integers, so are $6b - a$ and $7b$ and $7b \neq 0$;

hence $\sqrt{2}$ is a quotient of the two integers $6b - a$ and $7b$ with $7b \neq 0$.

Accordingly, $\sqrt{2}$ is rational (by definition of rational).

This contradicts the fact because $\sqrt{2}$ is irrational.

Hence our supposition is false and so $6 - 7\sqrt{2}$ is irrational.

(b) Solution:

Suppose $\sqrt{2} + \sqrt{3}$ is rational. Then, by definition of rational, there exists integers a and b with $b \neq 0$ such that

$$\sqrt{2} + \sqrt{3} = \frac{a}{b}$$

Squaring both sides, we get

$$2 + 3 + 2\sqrt{2}\sqrt{3} = \frac{a^2}{b^2}$$

$$\Rightarrow 2\sqrt{2 \times 3} = \frac{a^2}{b^2} - 5$$

$$\Rightarrow 2\sqrt{6} = \frac{a^2 - 5b^2}{b^2}$$

$$\Rightarrow \sqrt{6} = \frac{a^2 - 5b^2}{2b^2}$$

Since a and b are integers, so are therefore $a^2 - 5b^2$ and $2b^2$ with $2b^2 \neq 0$. Hence $\sqrt{6}$ is the quotient of two integers $a^2 - 5b^2$ and $2b^2$ with $2b^2 \neq 0$. Accordingly, $\sqrt{6}$ is rational. But this is a contradiction, since $\sqrt{6}$ is not rational. Hence our supposition is false and so $\sqrt{2} + \sqrt{3}$ is irrational.

REMARK:

The sum of two irrational numbers need not be irrational in general for

$$(6 - 7\sqrt{2}) + (6 + 7\sqrt{2}) = 6 + 6 = 12$$

which is rational.

(c) Solution:

Let z is any integer. [We must show that $4(z^2 + z + 1) - 3z^2$ is a perfect square.] Then, we have

$$4(z^2 + z + 1) - 3z^2 = 4z^2 + 4z + 4 - 3z^2 = z^2 + 4z + 4 = (z + 2)^2$$

Hence, it's a perfect square because $(z + 2)$ is an integer, being a sum of z and 2.

7. (a) Solution:

Basis step: $n=1$

$$n^2 = \frac{n(n+1)(2n+1)}{6} \quad 1 = \frac{1(1+1)(2+1)}{6} \quad 1 = \frac{6}{6} = 1 \quad 1 = 1 \quad \text{Hence } P(n) \text{ is true for } n=1.$$

Inductive step

$n = k$

$$P(k) = 1+2+3+\dots+k^2 = \frac{k(k+1)(2k+1)}{6}$$

$n = k+1$

$$\text{Since } P(k) = 1+2+3+\dots+k^2 = \frac{k(k+1)(2k+1)}{6}$$

Add $(k+1)^2$ both side

$$\begin{aligned} 1+2+3+\dots+k+(k+1) &= (k+1)^2 + \frac{k(k+1)(2k+1)}{6} \\ &= (k+1) \left(\frac{6(k+1)+k(2k+1)}{6} \right) \\ &= (k+1) \left(\frac{6k+6+2k^2+k}{6} \right) \\ &= (k+1) \left(\frac{2k^2+7k+6}{6} \right) = (k+1) \left(\frac{2k^2+4k+3k+6}{6} \right) \\ &= (k+1) \left(\frac{2k(k+2)+3(k+2)}{6} \right) = (k+1) \left(\frac{(2k+3)(k+2)}{6} \right) \\ &= \frac{(k+1)(k+2)(2k+3)}{6} \end{aligned}$$

Hence $p(n)$ is true for $n = k + 1$.

(b) Solution:

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{1}{4}n^2(n+1)^2$$

1. Show it is true for $n=1$

$$1^3 = \frac{1}{4} \times 1^2 \times 2^2 \text{ is True}$$

2. Assume it is true for $n=k$

$$1^3 + 2^3 + 3^3 + \dots + k^3 = \frac{1}{4}k^2(k+1)^2 \text{ is True (An assumption!)}$$

Now, prove it is true for " $k+1$ "

$$1^3 + 2^3 + 3^3 + \dots + (k+1)^3 = \frac{1}{4}(k+1)^2(k+2)^2$$

3. We know that $1^3 + 2^3 + 3^3 + \dots + k^3 = \frac{1}{4}k^2(k+1)^2$ (the assumption above), so we can do a replacement for all but the last term:

$$\frac{1}{4}k^2(k+1)^2 + (k+1)^3 = \frac{1}{4}(k+1)^2(k+2)^2$$

Multiply all terms by 4:

$$k^2(k+1)^2 + 4(k+1)^3 = (k+1)^2(k+2)^2$$

All terms have a common factor $(k+1)^2$, so it can be canceled:

$$k^2 + 4(k+1) = (k+2)^2$$

And simplify:

$$k^2 + 4k + 4 = k^2 + 4k + 4$$

They are the same! So it is true.

$$\text{So: } 1^3 + 2^3 + 3^3 + \dots + (k+1)^3 = \frac{1}{4}(k+1)^2(k+2)^2 \text{ is True.}$$

Rough work

$$\begin{aligned} n^2 &= \frac{n(n+1)(2n+1)}{6} \\ (k+1)^2 &= \left(\frac{(k+1)(k+2)(2k+3)}{6} \right) \end{aligned}$$

(c) Solution:

SOLUTION:

Let $P(n)$ be the given equation.

1. Basis Step:

$P(1)$ is true

For $n = 1$

$$\text{L.H.S of } P(1) = \frac{1}{1 \cdot 2} = \frac{1}{1 \times 2} = \frac{1}{2}$$

$$\text{R.H.S of } P(1) = \frac{1}{1+1} = \frac{1}{2}$$

Hence $P(1)$ is true

2. Inductive Step:

Suppose $P(k)$ is true, for some integer $k \geq 1$. That is

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{k(k+1)} = \frac{k}{k+1}$$

To prove $P(k+1)$ is true. That is

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{(k+1)(k+1+1)} = \frac{k+1}{(k+1)+1} \dots\dots\dots(2)$$

Now we will consider the L.H.S of the equation (2) and will try to get the R.H.S by using equation (1) and some simple computation.

Consider LHS of (2)

$$\begin{aligned} & \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{(k+1)(k+2)} \\ &= \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)} \\ &= \frac{k}{k+1} + \frac{1}{(k+1)(k+2)} \\ &= \frac{k(k+2)+1}{(k+1)(k+2)} \\ &= \frac{k^2+2k+1}{(k+1)(k+2)} \\ &= \frac{(k+1)^2}{(k+1)(k+2)} \\ &= \frac{k+1}{(k+2)} \\ &= \text{RHS of (2)} \end{aligned}$$

Hence $P(k+1)$ is also true and so by Mathematical induction the given equation is true for all integers $n \geq 1$.

8. (a)

(i) Solution: $q = 2$;

$r = 5$

(ii) Solution: $q = -11$;

$r = 10$

(iii) Solution: $q = 34$;

$r = 7$

(iv) Solution: $q = 77$;

$r = 0$

(v) Solution: $q = 0$;

$r = 10$

(vi) Solution: $q = 0$;

$r = 3$

(vii) Solution: $q = -1$;

$r = 2$

(viii) Solution: $q = 4$;

$r = 0$

(b)

$q = a \text{ div } m$

$r = a \text{ mod } m$

i) $a = -111, m = 99$.

Solution: $-2 = -111 \text{ div } 99$

$87 = -111 \text{ div } 99$

ii) $a = -9999, m = 101$.

Solution: $-99 = -9999 \text{ div } 101$

$0 = -9999 \text{ div } 101$

iii) $a = 10299, m = 999$.

Solution: $10 = 10299 \text{ div } 999$

$309 = 10299 \text{ div } 999$

iv) $a = 123456, m = 1001$.

Solution: $123 = 123456 \text{ div } 1001$

$333 = 123456 \text{ div } 1001$

(c)

i) Solution:

As We know that $a \equiv b \pmod{m}$ iff $\frac{a-b}{m}$.

Now $80 \not\equiv 5 \pmod{17}$ because $\frac{80-5}{17} = 4.41$.

ii) Solution:

As We know that $a \equiv b \pmod{m}$ iff $\frac{a-b}{m}$.

Now $103 \not\equiv 5 \pmod{17}$ because $\frac{103-5}{17} = 5.76$.

iii) Solution:

As We know that $a \equiv b \pmod{m}$ iff $\frac{a-b}{m}$.

Now $-29 \equiv 5 \pmod{17}$ because $\frac{-29-5}{17} = -2$.

iv) Solution:

As We know that $a \equiv b \pmod{m}$ iff $\frac{a-b}{m}$.

Now $-122 \not\equiv 5 \pmod{17}$ because $\frac{-122-5}{17} = -7.47$.

9. (a) (i) Solution: Yes $\gcd(11,15) = 1$, $\gcd(11,15) = 1$, $\gcd(11,19) = 1$
(ii) Solution: No $\gcd(14, 15) = 1$, $\gcd(14, 21) = 7$, $\gcd(15, 21) = 3$
iii) Solution: Yes. $\gcd(12, 17) = 1$, $\gcd(12, 31) = 1$, $\gcd(12, 37) = 1$, $\gcd(17, 31) = 1$, $\gcd(17, 37) = 1$, $\gcd(31, 37) = 1$
iv) Solution: Yes. $\gcd(7, 8) = 1$, $\gcd(7, 9) = 1$, $\gcd(7, 11) = 1$, $\gcd(8, 9) = 1$, $\gcd(8, 11) = 1$, $\gcd(9, 11) = 1$
(b)

i) Solution: $88 = 2^3 \cdot 11$

ii) Solution: $126 = 2 \cdot 3^2 \cdot 7$

iii) Solution: $729 = 3^6$

iv) Solution: $1001 = 7 \cdot 13 \cdot 11$

v) Solution: $1111 = 11 \cdot 101$

vi) Solution: $909 = 3^2 \cdot 101$

(c)

(i) Solution:

Prime factorization of the numbers: $11 = 11$; $15 = 3 \times 5$; $105 = 3 \times 5 \times 7$

LCM $(11, 15, 105) = 3 \times 5 \times 7 \times 11 = 1155$; $\text{GCD}(11, 15, 105) = 1$

(ii) Solution:

Prime factorization of the numbers: $14 = 2 \times 7$; $15 = 3 \times 5$; $21 = 3 \times 7$

LCM $(14, 15, 21) = 2 \times 3 \times 5 \times 7 = 210$; $\text{GCD}(14, 15, 21) = 1$

(iii) Solution:

Prime factorization of the numbers: $12 = 2 \times 2 \times 3$; $72 = 2 \times 2 \times 2 \times 3 \times 3$; $31 = 31$; $32 = 2 \times 2 \times 2 \times 2 \times 2$

LCM $(12, 72, 31, 32) = 2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 31 = 8928$; $\text{GCD}(12, 72, 31, 32) = 1$

(iv) Solution:

Prime factorization of the numbers: $17 = 17$; $18 = 2 \times 3 \times 3$; $192 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 3$; $1021 = 1021$

LCM $(17, 18, 192, 1021) = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 17 \times 1021 = 9997632$; $\text{GCD}(17, 18, 192, 1021) = 1$

10. (a)

(i) Solution: $\text{Gcd}(144,89) = (144)(34) + (89)(-55) = 1$

(ii) Solution: $\text{Gcd}(1001, 100001) = (10)(100001) + (-999)(1001) = 11$

(b)

(i) Solution:

$\text{Gcd}(55,89) = (55)(34) + (89)(-21) = 1$

So, inverse $\bar{a} = 34$.

Multiply 34 both side

$55 \cdot 34x \equiv 34 \cdot 34 \pmod{89}$

$x \equiv 1156 \pmod{89} = 88$.

(ii) Solution:

$\text{Gcd}(89,232) = (73)(89) + (232)(-28) = 1$

So, inverse $\bar{a} = 73$,

Multiply 73 both side

$89 \cdot 73x \equiv 2 \cdot 73 \pmod{232}$

$x \equiv 146 \pmod{232} = 146$.

(c)

(i) Solution: $\text{gcd}(2,17) = (1)(17) + (-8)(2) = 1$

So, $-8 + 17 = 9$

Hence inverse, $\bar{a} = 9$.

(ii) Solution: $\text{gcd}(34,89) = (13)(89) + (-34)(34) = 1$

So, $-34 + 89 = 55$

Hence inverse, $\bar{a} = 55$.

(iii) Solution: $\text{gcd}(144,233) = (89)(144) + (-55)(233) = 1$

Hence inverse, $\bar{a} = 89$.

(iv) Solution: $\text{gcd}(200,1001) = (1)(1001) + (-5)(200) = 1$

So, $-5 + 1001 = 996$

Hence inverse, $\bar{a} = 996$.

11. (a)

(i) Solution:

$X_{10} \equiv 1 \cdot 1 + 2 \cdot 2 + 3 \cdot 5 + 4 \cdot 9 + 5 \cdot 7 + 6 \cdot 3 + 7 \cdot 1 + 8 \cdot 2 + 9 \cdot 8 \pmod{11}$

$\equiv 1 + 4 + 15 + 36 + 35 + 18 + 7 + 16 + 72 \pmod{11} \equiv 204 \pmod{11}$

$X_{10} \equiv 6$. Yes, it's an original copy since $204 + (10 \cdot 6) \equiv 264 \pmod{11} \equiv 0$.

(ii) Solution:

$a = qd + r$

$26 = 3 \cdot 7 + 5$

$7 = 1 \cdot 5 + 2$

$5 = 2 \cdot 2 + 1$

Now, $1 = 1.5 - 2.2$;

$2 = 1.7 - 1.5$;

$5 = 1.26 - 3.7$

$$1 = 1.5 + 2(1.7 - 1.5) = 1.5 - 2.7 + 2.5; \quad 1 = 3.5 - 2.7 = 3(1.26 - 3.7) - 2.7$$

$$1 = 3.26 - 9.7 - 2.7 = 3.26 - 11.7; \quad 1 = (3)(26) + (-11)(7)$$

Inverse can't be negative so $-11 + 26 = 15$.

(b)

(i) Solution: E: $C = 04^3 \bmod 33$. X: $C = 23^3 \bmod 33$. A: $C = 00^3 \bmod 33$. M: $C = 12^3 \bmod 33$.

(ii) Solution: Encrypted Message: "STOP TALKING"

(c)

(i) Solution: -76, -51, -26, -1, 24, 49, 74, 99 are the integers.

(ii) Solution: By Fermat's little theorem, $5^{58} \equiv 1 \pmod{59}$, so $5^{116} \equiv 1 \pmod{59}$, so $5^{119} \equiv 5^3 \equiv 7 \pmod{59}$.

12. (a) (i) Solution:

We will follow the notation used in the proof of the Chinese remainder theorem.

We have $m = m_1 * m_2 * m_3 = 5 * 6 * 7 = 210$.

$M_1 = 210/5 = 42$, $M_2 = 210/6 = 35$, and $M_3 = 210/7 = 30$

Also, by simple inspection we see that:

$y_1 = 3$ is an inverse for $M_1 = 42$ modulo 5,

$y_2 = 5$ is an inverse for $M_2 = 35$ modulo 6 and

$y_3 = 4$ is an inverse for $M_3 = 30$ modulo 7.

The solutions to the system are then all numbers x such that $x = (a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3) \bmod m$
 $= ((1 * 42 * 3) + (2 * 35 * 5) + (3 * 30 * 4)) \bmod 210 = 836 \pmod{210} = 206$.

(ii) Solution:

We will follow the notation used in the proof of the Chinese remainder theorem.

We have $m = m_1 * m_2 * m_3 * m_4 = 2 * 3 * 5 * 11 = 330$.

$M_1 = 330/2 = 165$, $M_2 = 330/3 = 110$, $M_3 = 330/5 = 66$ and $M_4 = 330/11 = 30$

Also, by simple inspection we see that:

$y_1 = 1$ is an inverse for $M_1 = 165$ modulo 2, $y_2 = 2$ is an inverse for $M_2 = 110$ modulo 3,

$y_3 = 1$ is an inverse for $M_3 = 66$ modulo 5 and $y_4 = 7$ is an inverse for $M_4 = 30$ modulo 11.

The solutions to the system are then all numbers x such that $x = (a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3 + a_4 M_4 y_4) \bmod m$
 $= ((1 * 165 * 1) + (2 * 110 * 2) + (3 * 66 * 1) + (4 * 30 * 7)) \bmod 330 = 1643 \pmod{330} = 323$.

(b) Solution:

$x \equiv 3 \pmod{5}$, $x \equiv 3 \pmod{6}$, $x \equiv 1 \pmod{7}$, and $x \equiv 0 \pmod{11}$.

We will follow the notation used in the proof of the Chinese remainder theorem.

We have $m = m_1 * m_2 * m_3 * m_4 = 5 * 6 * 7 * 11 = 2310$.

Also, by simple inspection we see that:

$y_1 = 3$ is an inverse for $M_1 = 462$ modulo 5,

$y_2 = 1$ is an inverse for $M_2 = 385$ modulo 6,

$y_3 = 1$ is an inverse for $M_3 = 330$ modulo 7 and

$y_4 = 1$ is an inverse for $M_4 = 210$ modulo 11.

The solutions to the system are then all numbers x such that $x = (a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3 + a_4 M_4 y_4) \bmod m$
 $= (3 * 462 * 3) + (3 * 385 * 1) + (1 * 330 * 1) + (0 * 210 * 1) = 5643 \pmod{2310} = 1023$.

He could have 1023 oranges.

(c) Solution:

$x \equiv 3 \pmod{4}$, $x \equiv 5 \pmod{9}$, and $x \equiv 8 \pmod{13}$

We will follow the notation used in the proof of the Chinese remainder theorem.

We have $m = m_1 * m_2 * m_3 = 4 * 9 * 13 = 468$.

Also, by simple inspection we see that:

$y_1 = 1$ is an inverse for $M_1 = 117$ modulo 4,

$y_2 = 7$ is an inverse for $M_2 = 52$ modulo 9,

$y_3 = 4$ is an inverse for $M_3 = 36$ modulo 13.

The solutions to the system are then all numbers x such that $x = (a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3) \bmod m$
 $= (3 * 117 * 1) + (5 * 52 * 7) + (8 * 36 * 4) = 2543 \pmod{468} = 203$. He has received 203 presents.

13. (a) (i) Solution: The encrypted message will be "L ORYH GLVFUWHW PDWKHPDWLFV"

(ii) (i) Solution:

S T O P P O L L U T I O N
 18 19 14 15 15 14 11 11 20 19 8 14 13

After applying function:

22 23 18 19 19 18 15 15 24 23 12 18 17

W X S T T S P P Y X M S R

will be encrypted message.

ii) Solution:

S T O P P O L L U T I O N
18 19 14 15 15 14 11 11 20 19 8 14 13

After applying function:

13 14 09 10 10 09 06 06 15 14 03 09 08
N O J K K J G G P O D J I

will be encrypted message.

(b) (i) Solution: "MID TWO ASSIGNMENT" will be decrypted message.

(ii) Solution: "FAST NUCES UNIVERSITY" will be decrypted message.

(c) (i) Solution: "SURRENDER NOW" will be decrypted message.

(ii) Solution: "BE MY FRIEND" will be decrypted message.

14. (a)

(i) Solution: $034567981 \bmod 97 = 91$

(ii) Solution: $183211232 \bmod 97 = 57$

(iii) Solution: $220195744 \bmod 97 = 21$

(iv) Solution: $987255335 \bmod 97 = 5$

(b)

(i) Solution: $104578690 \bmod 101 = 58$.

(ii) Solution: $432222187 \bmod 101 = 60$.

(iii) Solution: $372201919 \bmod 101 = 32$.

(iv) Solution: $501338753 \bmod 101 = 3$.

(c)

(i) Solution:

$$X_1 = (4 * 3 + 1) \bmod 7 = 6.$$

$$X_2 = (4 * 6 + 1) \bmod 7 = 4.$$

$$X_3 = (4 * 4 + 1) \bmod 7 = 3.$$

$$X_4 = (4 * 3 + 1) \bmod 7 = 6.$$

$$X_5 = (4 * 6 + 1) \bmod 7 = 4 \quad \text{Sequence: } 6, 4, 3, 6, 4, \dots$$

(ii) Solution:

$$X_1 = (7 * 5 + 2) \bmod 9 = 1.$$

$$X_2 = (7 * 1 + 2) \bmod 9 = 0.$$

$$X_3 = (7 * 0 + 2) \bmod 9 = 2.$$

$$X_4 = (7 * 2 + 2) \bmod 9 = 7.$$

$$X_5 = (7 * 7 + 2) \bmod 9 = 6 \quad \text{Sequence: } 1, 0, 2, 7, 6, \dots$$

15. (a) (i) Solution:

$$7*3 + 3 + 2*3 + 3 + 2*3 + 1 + 8*3 + 4 + 4*3 + 3 + 4*3 + x_{12} = 0 \bmod 10$$

$$21 + 3 + 6 + 3 + 6 + 1 + 24 + 4 + 12 + 3 + 12 + x_{12} = 0 \bmod 10$$

$$95 + x_{12} = 0 \bmod 10$$

Check digit is $x_{12} = 5$.

(ii) Solution:

$$6*3 + 3 + 6*3 + 2 + 3*3 + 9 + 9*3 + 1 + 3*3 + 4 + 6*3 + x_{12} = 0 \bmod 10$$

$$18 + 3 + 18 + 2 + 9 + 9 + 27 + 1 + 9 + 4 + 18 + x_{12} = 0 \bmod 10$$

$$118 + x_{12} = 0 \bmod 10$$

Check digit is $x_{12} = 2$.

(b) (i) Solution:

$$0*3 + 3 + 6*3 + 0 + 0*3 + 0 + 2*3 + 9 + 1*3 + 4 + 5*3 + 2 = 0 \bmod 10$$

$$0 + 3 + 18 + 0 + 0 + 0 + 6 + 9 + 3 + 4 + 15 + 2 = 0 \bmod 10$$

$$60 \equiv 0 \bmod 10$$

It's a valid UPC code.

(ii) Solution:

$$0*3 + 1 + 2*3 + 3 + 4*3 + 5 + 6*3 + 7 + 8*3 + 9 + 0*3 + 3 = 0 \bmod 10$$

$$0 + 1 + 6 + 3 + 12 + 5 + 18 + 7 + 24 + 9 + 0 + 3 = 0 \bmod 10$$

$$88 \not\equiv 0 \bmod 10$$

It's not a valid UPC code.

(c) (i) Solution:

$$1*0 + 2*0 + 3*7 + 4*1 + 5*1 + 6*9 + 7*8 + 8*8 + 9*1 + x_{10} = 0 \bmod 11$$

$$0 + 0 + 21 + 4 + 5 + 54 + 56 + 64 + 9 + x_{10} = 0 \bmod 11$$

$$213 + x_{10} = 0 \bmod 11$$

Check digit, $x_{10} = 4$.

(ii)Solution:

$$\begin{aligned}x_{10} &= 1 \cdot 0 + 2 \cdot 3 + 3 \cdot 2 + 4 \cdot 1 + 5 \cdot 5 + 6 \cdot 0 + 7 \cdot 0 + 8 \cdot Q + 9 \cdot 1 \pmod{11} \\&= 0 + 6 + 6 + 4 + 25 + 0 + 0 + 8Q + 9 \pmod{11} \\&= 8Q + 50 \pmod{11}\end{aligned}$$

The check digit is known to be 8.

$$8Q + 50 \pmod{11} = 8$$

Since $50 \pmod{11} = 6$

$$8Q + 6 \pmod{11} = 8$$

Subtract 6 from each side of the equation:

$$8Q \pmod{11} = 2$$

Since the inverse of $8 \pmod{11}$ is $7 \pmod{11}$, we should multiply both sides of the equation by 7:

$$\begin{aligned}7 \cdot 8Q \pmod{11} &= 7 \cdot 2 \pmod{11} \\56Q \pmod{11} &= 14 \pmod{11} \\Q \pmod{11} &= 3\end{aligned}$$

Since Q is a digit (between 0 and 9), Q then has to be equal to 3.

16. (a) Solution:

A T T A C K
00 19 19 00 02 10

• $n = 43 \cdot 59 = 2537$; $e = 13$ Encryption Function: $C = M^e \pmod{n}$
 $C(AT) = 0019^{13} \pmod{2537}$ $C(TA) = 1900^{13} \pmod{2537}$ $C(CK) = 0210^{13} \pmod{2537}$

E X A M I N A T I O N
04 23 00 12 08 13 00 19 08 14 13

• $n = 43 \cdot 59 = 2537$; $e = 13$ Encryption Function: $C = M^e \pmod{n}$
 $C(EX) = 0423^{13} \pmod{2537}$ $C(AM) = 0012^{13} \pmod{2537}$ $C(IN) = 0813^{13} \pmod{2537}$
 $C(AT) = 0019^{13} \pmod{2537}$ $C(IO) = 0814^{13} \pmod{2537}$ $C(N) = 0013^{13} \pmod{2537}$

(b) Solution:

A S S I G N M E N T
00 18 18 08 06 13 12 04 13 19

• $n = 13 \cdot 19 = 247$ $e = 47$ Encryption Function: $C = M^e \pmod{n}$
 $C(A) = 00^{47} \pmod{247}$ $C(S) = 18^{47} \pmod{247}$ $C(I) = 08^{47} \pmod{247}$ $C(G) = 06^{47} \pmod{247}$
 $C(N) = 13^{47} \pmod{247}$ $C(M) = 12^{47} \pmod{247}$ $C(E) = 04^{47} \pmod{247}$ $C(T) = 19^{47} \pmod{247}$

S E M E S T E R
18 04 12 04 18 19 04 17

• $n = 13 \cdot 19 = 247$ $e = 47$ Encryption Function: $C = M^e \pmod{n}$
 $C(S) = 18^{47} \pmod{247}$ $C(E) = 04^{47} \pmod{247}$ $C(M) = 12^{47} \pmod{247}$ $C(T) = 19^{47} \pmod{247}$ $C(R) = 17^{47} \pmod{247}$

(c) (i) Solution: Since $5^6 = 1 \pmod{7}$; $(5^6)^{333} \cdot 5^5 \pmod{7} = 5^5 \pmod{7} = 3$.

(ii) Solution: Since $5^{10} = 1 \pmod{11}$; $(5^{10})^{2000} \cdot 5^3 \pmod{11} = 5^3 \pmod{11} = 4$.

(iii) Solution: Since $5^{12} = 1 \pmod{13}$; $(5^{12})^{166} \cdot 5^{11} \pmod{13} = 5^{11} \pmod{13} = 8$.

(iv) Solution: Since $5^{16} = 1 \pmod{17}$; $(5^{16})^{126} \cdot 5^3 \pmod{17} = 5^3 \pmod{17} = 6$.

17. (a) Solution:

We will follow the notation used in the proof of the Chinese remainder theorem.

We have $m = m_1 \cdot m_2 \cdot m_3 = 2261$.

Also, by simple inspection we see that:

$y_1 = 1$ is an inverse for $M_1 = 323$ modulo 7,

$y_2 = 11$ is an inverse for $M_2 = 133$ modulo 17, and

$y_3 = 4$ is an inverse for $M_3 = 119$ modulo 19.

The solutions to the system for all numbers x such that $x = (a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3) \pmod{m}$

$x = ((2 \cdot 323 \cdot 1) + (3 \cdot 133 \cdot 11) + (5 \cdot 119 \cdot 4)) \pmod{2261} = 7415 \pmod{2261} = 632$. He would have 632 balls.

(b) Solution: six hundred and thirty two

$$f(p) = (p + 3) \bmod 26$$

s i x h u n d r e d a n d t h i r t y t w o
18 8 23 7 20 13 3 17 4 3 0 13 3 19 7 8 17 19 24 19 22 14

Encrypted Text: VLA KXQGUHG DQG WKLUWB WZR

(c) Solution: Number = 632 so $x = 6 + 3 + 2 = 11$.

By Fermat's little theorem, we know that $11^{10} \equiv 1 \pmod{7}$, and so $(11^{10})^k \equiv 1 \pmod{7}$, for every positive integer k .
Therefore,

$$11^{302} = 11^{30 \cdot 10 + 2} = (11^{10})^{30} \cdot 11^2 \equiv (1)^{30} \cdot 121 \equiv 2 \pmod{7}. \quad \text{Hence, } 11^{302} \bmod 7 = 2.$$

18. (a) Solution: There are $27 \cdot 37 = 99$ offices in the building.

(b) Solution: $12 \cdot 2 \cdot 3$ shirts are required.

(c) Solution: People can have $26 \cdot 26 \cdot 26 = 26^3$ different three-letter initials.

19. (a) Solution: There are ${}^{13}C_5 = 1287$ ways to select the players.

(b) Solution: There are ways ${}^{75}C_5 = 2071126800$ to place the people on different positions.

(c) Solution: There are ${}^{27}C_5 = 17,550$ combinations to take the ride.

20. (a) Solution: There are 16 place values for hexadecimal numbers: 0 to 9, A, B, C, D, E and F.

So, $16^{10} + 16^{28} + 16^{58}$ different WEP keys are possible.

(b) Solution: There would be $26^4 - 25^4 = 66,351$ strings.

(c) Solution: People can have $26 \cdot 25 \cdot 24 = 15,600$ different three-letter initials with none of the letters repeated.

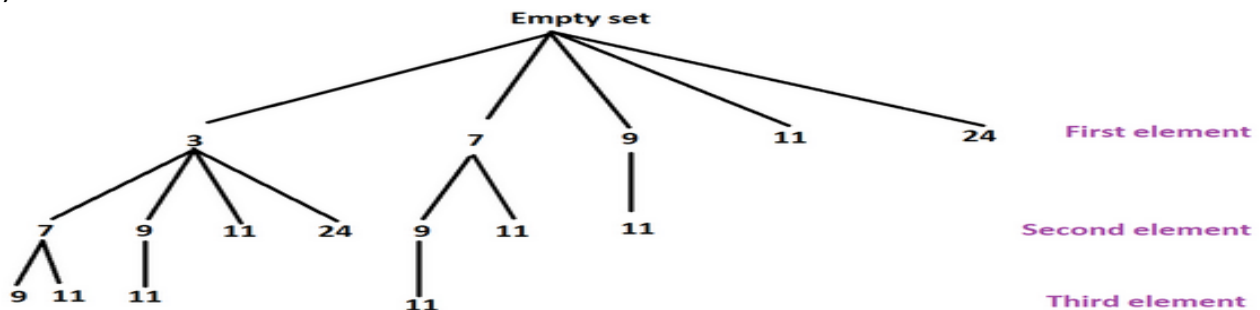
21. (a) Solution: Since each value of the domain can be mapped to one of two values.

Number of functions are: $2 \cdot 2 \cdot 2 \cdot 2 \dots \cdot m = 2^m$.

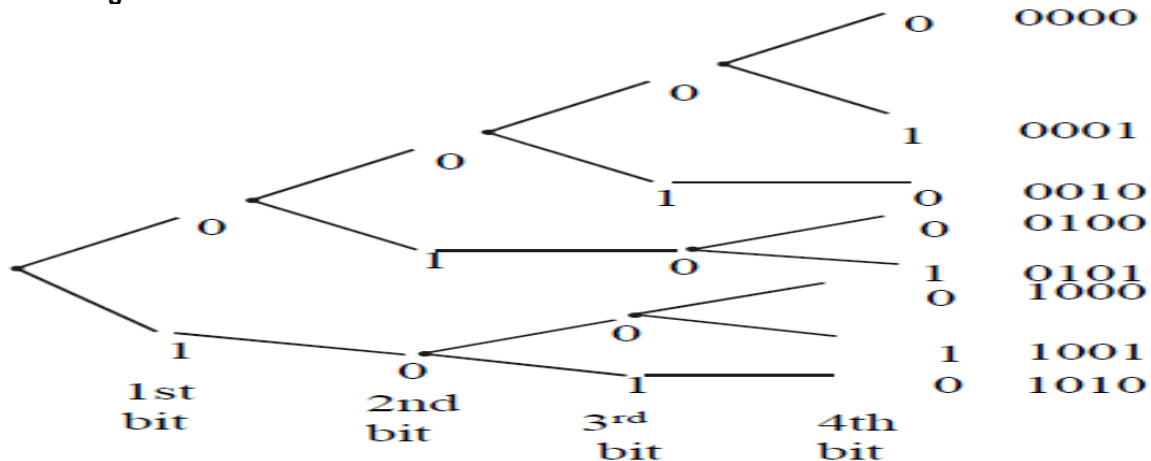
(b) Solution: Each successive element from the domain will have one option than its predecessor as it is one-to-one function. So, number of functions are $5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$.

(c) Solution: There are $15 \cdot 48 \cdot 24 \cdot 34 \cdot 28 \cdot 28 = 460,615,680$ different faces.

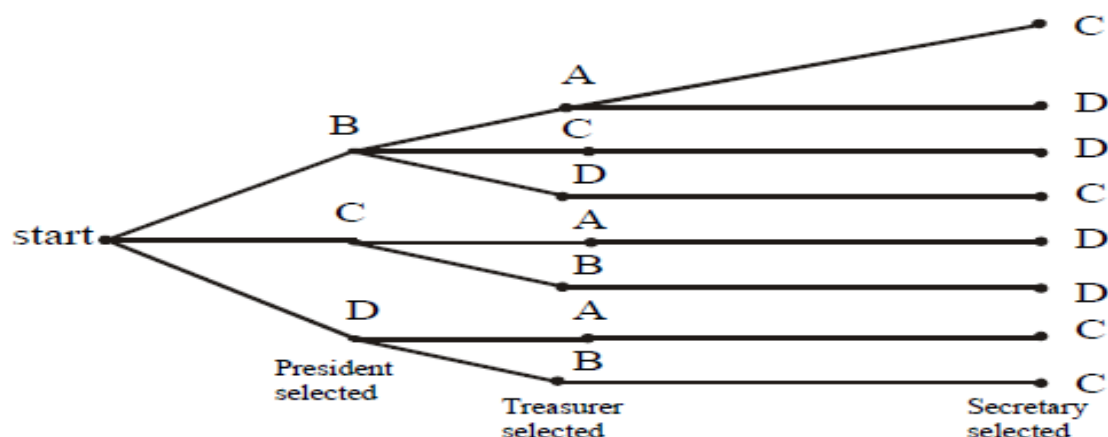
22. (a) Solution:



(b) Solution: The following tree diagrams displays all bit strings of length four without two consecutive 1's. Clearly, there are 8 bit strings.



(c) Solution: We construct the possibility tree to see all the possible choices. From the tree given below, see that there are only eight ways possible to choose the offices under given conditions.



23. (a) Solution: There are ${}^8C_3 = 56$ ways to choose the students.
 (b) Solution: There are ${}^{12}C_6 = 924$ ways to select the elective courses.
 (c) Solution: ${}^9C_5 = 126$ different teams can be selected.
24. (a) Solution: There are ${}^{20}P_5 = 1,860,480$ ways to choose a chairperson, assistant chairperson, treasurer, community advisor, and record keeper.
 (b) Solution: There are ${}^{16}P_4 = 43,680$ ways coach can select students to compete in the race.
 (c) Solution: There are ${}^{15}P_2 = 210$ ways student can be chosen for these 2 positions.
25. (a) Solution: ${}^5C_1 * {}^3C_2 * {}^4C_1 * {}^6C_3 = 1200$ Sandwiches can be made of 1 meat, 2 bread, 1 cheese, and 3 condiments.
 (b) Solution: There are ${}^{12}C_5 = 792$ different toppings on ice cream.
 (c)
 (i) Solution: With repetition: $1*10*10*10*10 = 10,000$ Zip codes Without repetition: $1*9*8*7*6 = 3024$ Zip codes
 (ii) Solution: With repetition: $2*10*10*10*10 = 20,000$ Zip codes Without repetition: $2*9*8*7*6 = 6048$ Zip codes
26. (a) Solution:
 $A = \text{Strings begins with three 0s} = 2^7 = 128$ $B = \text{Strings end with two 0s} = 2^8 = 256$ $A \cap B = 2^5 = 32$
 $A \cup B = A + B - A \cap B = 128 + 256 - 32 = 352$.
 (b) Solution:
 $A = \text{Strings begins with 0s} = 2^4 = 16$ $B = \text{Strings end with two 1s} = 2^3 = 8$ $A \cap B = 2^2 = 4$
 $A \cup B = A + B - A \cap B = 16 + 8 - 4 = 20$.
 (c)
 (i) Solution: With repetition: $26*26*10*10*10*10 = 6,760,000$ Without repetition: $26*25*10*9*8*7 = 3,276,000$
 (ii) Solution: With repetition: $5*5*5*5*5 = 3125$ Without repetition: $5*4*3*5*4 = 1200$
27. (a) Solution: The first letter of each last name are the pigeonholes, and the letters of the alphabet are pigeons. By the generalized pigeonhole principle, $\lceil 30/26 \rceil = 2$. So, there are at least two students, have last names that begin with the same letter.
 (b) Solution: By the generalized pigeonhole principle, $\lceil 8008278/1000000 \rceil = 9$.
 (c) Solution: The 38 time periods are the pigeonholes, and the 677 classes are the pigeons. By the generalized pigeonhole principle there is at least one time period in which at least $\lceil 677/38 \rceil = 18$ classes are meeting. Since each class must meet in a different room, we need 18 rooms.
28. (a) (i) Solution: From binomial theorem, it follows that coefficient is: ${}^nC_r = {}^{11}C_5 = 462$.
 (ii) Solution: From binomial theorem, it follows that coefficient is: ${}^nC_r = {}^{24}C_{17} (2)^7 (-1)^{17} = -44,301,312$.

(b) (i) Solution:

$$\begin{array}{ccccccc}
 & & 1 & & & & \leftarrow \text{row 0} \\
 & 1 & & 1 & & & \\
 & & 1 & & 2 & & 1 \\
 & 1 & & 3 & & 3 & & 1 \\
 & 1 & & 4 & & 6 & & 4 & & 1 \leftarrow \text{row 4} \\
 & 1 & & 5 & & 10 & & 10 & & 5 & & 1 \leftarrow \text{row 5} \\
 & 1 & & 6 & & 15 & & 20 & & 15 & & 6 & & 1 \leftarrow \text{row 6}
 \end{array}$$

$$\begin{array}{c}
 \binom{0}{0} \\
 \binom{1}{0} \quad \binom{1}{1} \\
 \binom{2}{0} \quad \binom{2}{1} \quad \binom{2}{2} \\
 \binom{3}{0} \quad \binom{3}{1} \quad \binom{3}{2} \quad \binom{3}{3} \\
 \binom{4}{0} \quad \binom{4}{1} \quad \binom{4}{2} \quad \binom{4}{3} \quad \binom{4}{4} \\
 \binom{5}{0} \quad \binom{5}{1} \quad \binom{5}{2} \quad \binom{5}{3} \quad \binom{5}{4} \quad \binom{5}{5} \\
 \binom{6}{0} \quad \binom{6}{1} \quad \binom{6}{2} \quad \binom{6}{3} \quad \binom{6}{4} \quad \binom{6}{5} \quad \binom{6}{6}
 \end{array}$$

(ii) Solution:

$$= 1(3x)^4(-2)^0 + 4(3x)^3(-2)^1 + 6(3x)^2(-2)^2 + 4(3x)^1(-2)^3 + 1(3x)^0(-2)^4$$

$$= 3^4 x^4 + 4 \cdot 3^3 \cdot (-2) x^3 + 6 \cdot 3^2 \cdot 4 x^2 + 4 \cdot 3 \cdot (-8) x + 16$$

$$= 81x^4 - 216x^3 + 216x^2 - 96x + 16$$

$$\begin{array}{ccccccc}
 & & & & 1 & & \\
 & & & 1 & & 1 & \\
 & & 1 & & 2 & & 1 \\
 & 1 & & 3 & & 3 & & 1 \\
 1 & & 4 & & 6 & & 4 & & 1
 \end{array}$$

(c) (i) Solution:

$$= {}^6C_0(2x)^{6-0}(y)^0 + {}^6C_1(2x)^{6-1}(y)^1 + {}^6C_2(2x)^{6-2}(y)^2 + {}^6C_3(2x)^{6-3}(y)^3 + {}^6C_4(2x)^{6-4}(y)^4 + {}^6C_5(2x)^{6-5}(y)^5 + {}^6C_6(2x)^{6-6}(y)^6$$

$$(2x+y)^6 = 64x^6 + 192x^5y + 240x^4y^2 + 160x^3y^3 + 60x^2y^4 + 12xy^5 + y^6$$

(ii) Solution:

$$= {}^7C_0(3a)^{7-0}(-2b)^0 + {}^7C_1(3a)^{7-1}(-2b)^1 + {}^7C_2(3a)^{7-2}(-2b)^2 + {}^7C_3(3a)^{7-3}(-2b)^3 + {}^7C_4(3a)^{7-4}(-2b)^4 + {}^7C_5(3a)^{7-5}(-2b)^5 + {}^7C_6(3a)^{7-6}(-2b)^6 + {}^7C_7(3a)^{7-7}(-2b)^7$$

$$= (3a - 2b)^7 = 2187a^7 - 10206a^6b + 20412a^5b^2 - 22680a^4b^3 + 15120a^3b^4 - 6048a^2b^5 + 1344ab^6 - 128b^7$$

29. (a) (i) Solution: There are 36 students. They can be put in a row in 36! ways.

(ii) Solution: You need to have an ordered arrangement of 7 out of 36 students. The number of such arrangements is $P(36, 7)$.

(iii) Solution: You need to have an ordered arrangement of all 20 women and ordered arrangement of all 16 men. By the product rule, this can be done in $20! \cdot 16!$ ways.

(b) Solution: No pitcher * Last 5 pitcher in = ${}^8P_4 \cdot {}^5P_5 = 201,600$.

(c) Solution: ${}^{12}C_0 + {}^{12}C_1 + {}^{12}C_2 + {}^{12}C_3 = 299$.

30. (a) Solution: Expression = $({}^{15}C_3) ({}^{12}C_3) ({}^7C_3) ({}^5C_3) ({}^9C_3) ({}^{10}C_3) = 3.531 \times 10^{11}$.

(b) Solution:

Let N be 97 students and K be 10 different grades.

$\lceil N/K \rceil = \lceil 97/10 \rceil = 10$. So, students 10 will earn the exact same grade.

(c) Solution: Squad is consisting of 15 players so, ${}^{15}P_1 \times {}^{14}P_1 \times {}^{13}P_1 = 420$.